



# pyGIMLi

*Geophysical Inversion & Modelling Library*

*A short course on versatile processing & inversion of ERT monitoring data*

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<https://gelmon25.pygimli.org>

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# introducing pyGIMLi

is rather a box full of tools and not a specific tool

aims at making:

- easy things easy
- hard things possible
- everything transparent and reproducible

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Pingo!



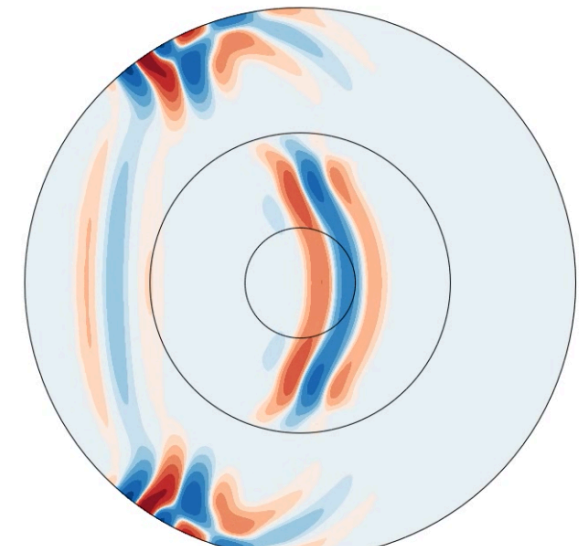
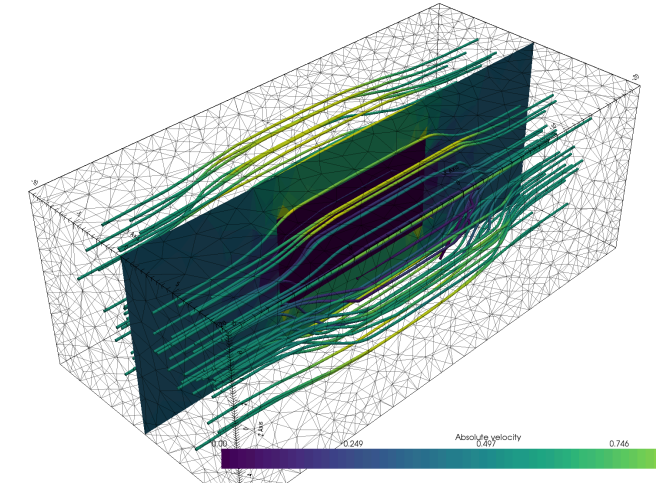
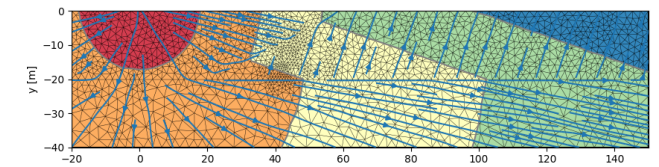
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 **pyGIMLi** is a versatile open-source toolbox with:

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- management tools for **structured and unstructured meshes** in 2D & 3D
- computationally efficient **finite-element and finite-volume solvers**
- **various geophysical forward operators**: ERT/IP, Traveltime, Gravimetry, Magnetics, SP, EM
- frameworks for **constrained, joint and process-based inversions** with **region-specific regularization**
- **open-source, platform compatible**, documented & tested code
- suitability for **teaching & reproducible research**
- v1.0 published in *Computers and Geosciences* (Rücker et al., 2017) (among *Most Downloaded* papers, > 430 citations, > 120 uses in peer-reviewed papers)



# ◀ Existing tutorials

- Transform 2021: creating **geometries & meshes**, **modeling** PDEs, **synthetic data**, **inversion** (also with **external forward operator**).
- Transform 2022: fundamental **pyGIMLi** objects (**Mesh**, **DataContainer**, matrix types, etc.), **geostatistical** vs. smoothness regularization, treatment of subsurface **regions**, adding **prior data**.
- SEG webinar 2024: invert **real-life 3D data** (Hübner et al., 2017) to tweak your **inversion beyond the standard** practice.
- Today we **practise** & focus on **ERT monitoring** data processing & inversion.

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The collage features several YouTube video thumbnails and titles:

- Tutorial: Geophysical modeling & inversion with pyGIMLi** (11.671 Aufrufe • vor 2 Jahren gestreamt) by Software Underground. The thumbnail shows a 2D cross-section plot with a green and yellow region.
- Tutorial: PyGIMLi** (3284 Aufrufe • vor 1 Jahr gestreamt) by Software Underground. The thumbnail shows a 3D visualization of a subsurface model with red and blue regions.
- Setting up PyGimli** (1178 Aufrufe • vor 2 Jahren) by Brady Flinchum. The thumbnail shows a terminal window with code.
- Open-source hydrogeophysical modeling and inversion with pyGIMLi 1.1** (1080 Aufrufe • vor 3 Jahren) by Florian Wagner. The thumbnail shows a presentation slide with logos of various institutions.

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## What is new since?

- Transform '21 & '22 notebooks available as **tutorials** or **examples** on [pygimli.org](https://pygimli.org)
- **Improved 3D visualization** powered by [pyvista](https://pyvista.org) (filters, slices, interactivity)
- **3D gravity and (full-tensor) magnetics** operators and managers
- **New matrices and matrix generators**, e.g. non-explicit (PDE-based) Jacobian matrix
- [LSQRinversion](#) framework to include parameter relations (from [Wagner et al., 2019](#))
- [MultiFrameModelling](#) framework for temporally/spectrally/spatially constraints
- [TimelapseERT](#) class with different strategies, e.g. **4D inversion**
- **New examples** on ERT (2D/3D crosshole, 3D surface, timelapse), IP, 3D magnetics
- Many more **convenience functions** to simplify the code
- Many **new papers using pyGIMLi** (<https://pygimli.org/publist.html>)

## New since v1.5.0 (SEG webinar)

<https://github.com/gimli-org/gimli/releases>

- enabled `pip install pygimli` for easier installation (e.g. on Google Colab)
- new **examples** (e.g. reciprocals) and improved old ones
- better access to **Jacobian, resolution matrices** etc.
- **syntactic sugar** (`data.show('i')`, `mesh.show('res')`, `data.estimateError()`)
- new **mesh functions**:  
e.g., `submesh()`, `extract2dUpperSurfaceMesh()`, `extract2dSlice()`
- new **matrix types**, e.g. for BFGS-based inversion
- different **inversion frameworks** (SD, NLCG, GN, L\_BFGS, BFGS)
- fully **complex-valued** (FD) ERT-IP inversion (also for TD)
- improvements in **gravity, magnetics and TDEM**
- a lot of bug fixes and improved docstrings

# 👥 Join the pyGIMLi user community!



“In open source, we feel strongly that to really **do something well**, you have to get a lot of people involved.”

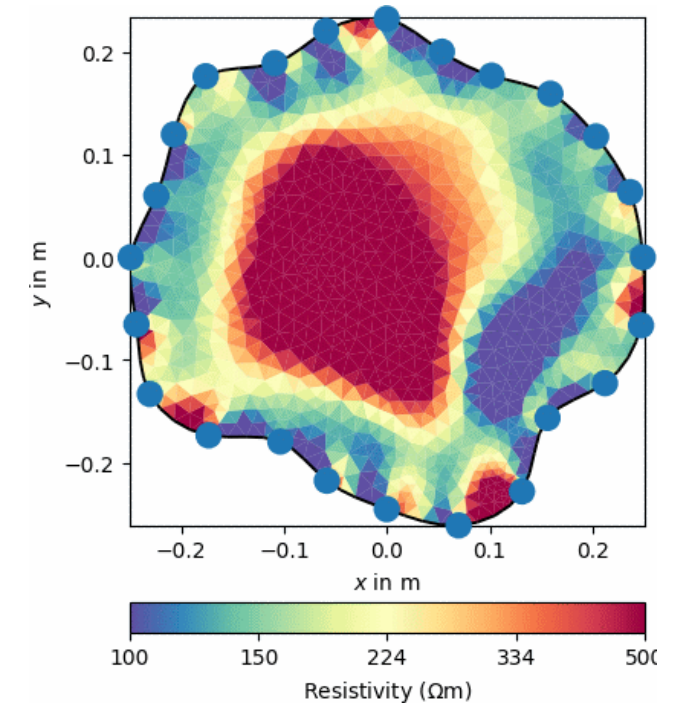
– *Linus Torvalds*

1. Join the **#pyGIMLi** chat on Mattermost!
2. **Open a discussion** or **raise an issue** on  GitHub.
3. **Contribute to the website** via the “Improve this page” button in the right sidebar.
4. **Add your** pyGIMLi-powered **publication** to [this database](#).
5. **Send your example** to [mail@pygimli.org](mailto:mail@pygimli.org).
6. **Contribute to the code** as described in [our contribution guidelines](#).



# The code base – ERT module

- import and export of various formats
- utility functions and data visualization
- array generation and synthetic model simulation
- **ERTManager** for data inversion and model output
- **ERTIPManager** for FD or TD Induced Polarization (IP)
- **TimelapseERT** (and **CrossholeERT**) class for timelapse ERT
- spectral IP analysis in **pyBERT** module (**FDIP** and **TDIP**)





## </> The `TimelapseERT` class

- designed for surface ERT data (crosshole: `CrossholeERT`)
- different import formats and export of filtered data
- extract data time series with `.showTimeline()`
- extensive filter function (`t`, `tmin`, `tmax`, `kmax`)
- masking of data with app. resistivity & error bounds (`.mask()`)
- automatic masking based on harmonic filtering
- different types of data visualization and PDF generation
- temporal fitting of reciprocal error models
- invert single timesteps, sequentially or full (“4D” inversion)

## </> The CrossholeERT class

- based on `TimelapseERT` with some extra functions
- different visualization methods
- adapted mesh generation
- attributes electrodes to boreholes
- `extractSubset` to retrieve (3D) parts or (2D) planes

</> Make whole processing chain transparent and reproducible

## ☒ TimelapseERT repository

Github Repository to collect published data sets and notebooks

| Name         | Reference              | dim   | time | data | Info              |
|--------------|------------------------|-------|------|------|-------------------|
| pyGIMLi      | (Rücker et al., 2017)  | 2D    | 10   | 740  | synthetic tracer  |
| ALERT        | (Kuras et al., 2009)   | 2D xh | 35   | 1200 | 9 boreholes       |
| Hillslope    | (Hübner et al., 2015)  | 2D to | 24   | 800  | min. length       |
| Infiltration | (Hübner et al., 2017)  | 3D    | 200  | 2850 | min. length       |
| Tsunami      | (Ronczka et al., 2014) | 2.5D  | 900  | 225  | lab experiment    |
| Steelcase    | (Ronczka et al., 2015) | 3D sc | 9    | 450  | SEM, regions      |
| SAMOS        | (Ronczka et al., 2020) | 1.5D  | 7    | 3000 | CEM               |
| DynaDeep     | Skibbe talk            | 2D to | 10   | 3000 | topography change |
| TestUM       | Günther talk           | 3D xh | 70   | 2200 | heat experiment   |

# How to get started

1. Open **Miniforge Prompt** (🪟) or a Terminal (🍏/🐧).
2. Clone the **GELMON25** repository.

```
git clone https://github.com/gimli-org/gelmon25.git
cd gelmon25
```

3. Install the **pg** environment with required dependencies (particularly **pygimli=1.5.3**).

```
conda env create
```

4. Activate environment and start a **Jupyter Notebook**.

```
conda activate pg
jupyter notebook
```



## Follow without a local installation

You can also visit <https://colab.research.google.com>, open an empty notebook and type **!pip install pygimli tetgen**.

# Practising

▶▶ Let's find out what you already know and what you're interested in!

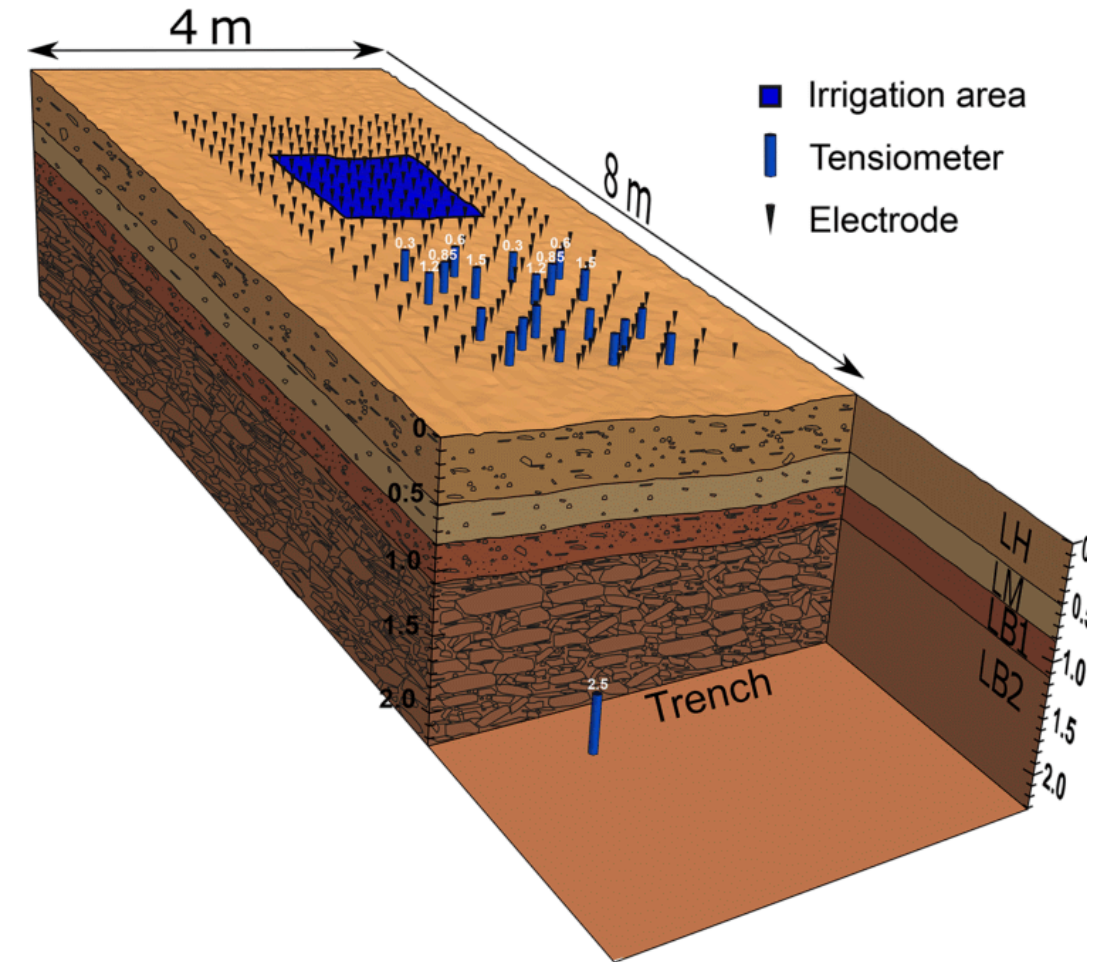
<https://gelmon25.pygimli.org>

# Do you know it and if no do you want to learn it?

- array generation and sensitivity distribution
- mesh generation and synthetic modelling
- field data inversion
- error estimation and reciprocal data analysis
- structural constraints in
- using petrophysical relations
- a-priori constraints
- image appraisal (resolution and DOI)
- basic timelapse strategies
- automated timelapse processing
- complex conductivity inversion
- spectral (FD or TD) inversion of DC/IP data

## ►► The infiltration case (s. also SEG webinar)

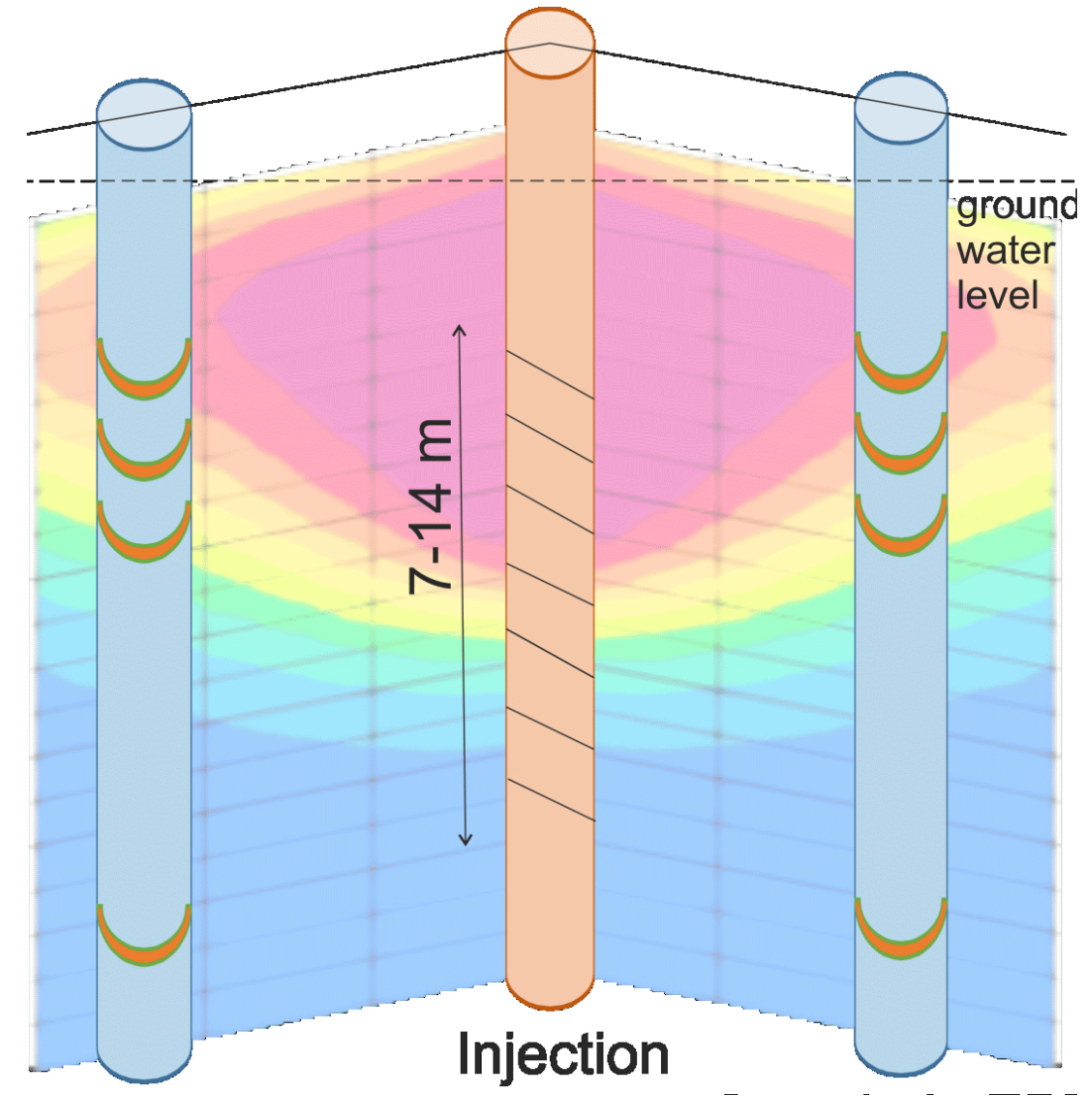
1. Load, process, visualize data
2. Extract 2D profiles from 3D
3. Test different meshes
4. Different regularizations
5. Visualize results in 2D & 3D
6. Invert for saturation
7. Add prior information
8. Time-lapse inversion



Survey layout after Hübner et al. (2017)

## ►► The heat case (s. also talk)

1. Load, and visualize data
2. look at data timeseries
3. Filter and mask data
4. select 2D subsets
5. Invert single frames
6. Time-lapse inversion
7. Visualize results in 2D & 3D
8. Transfer to temperature

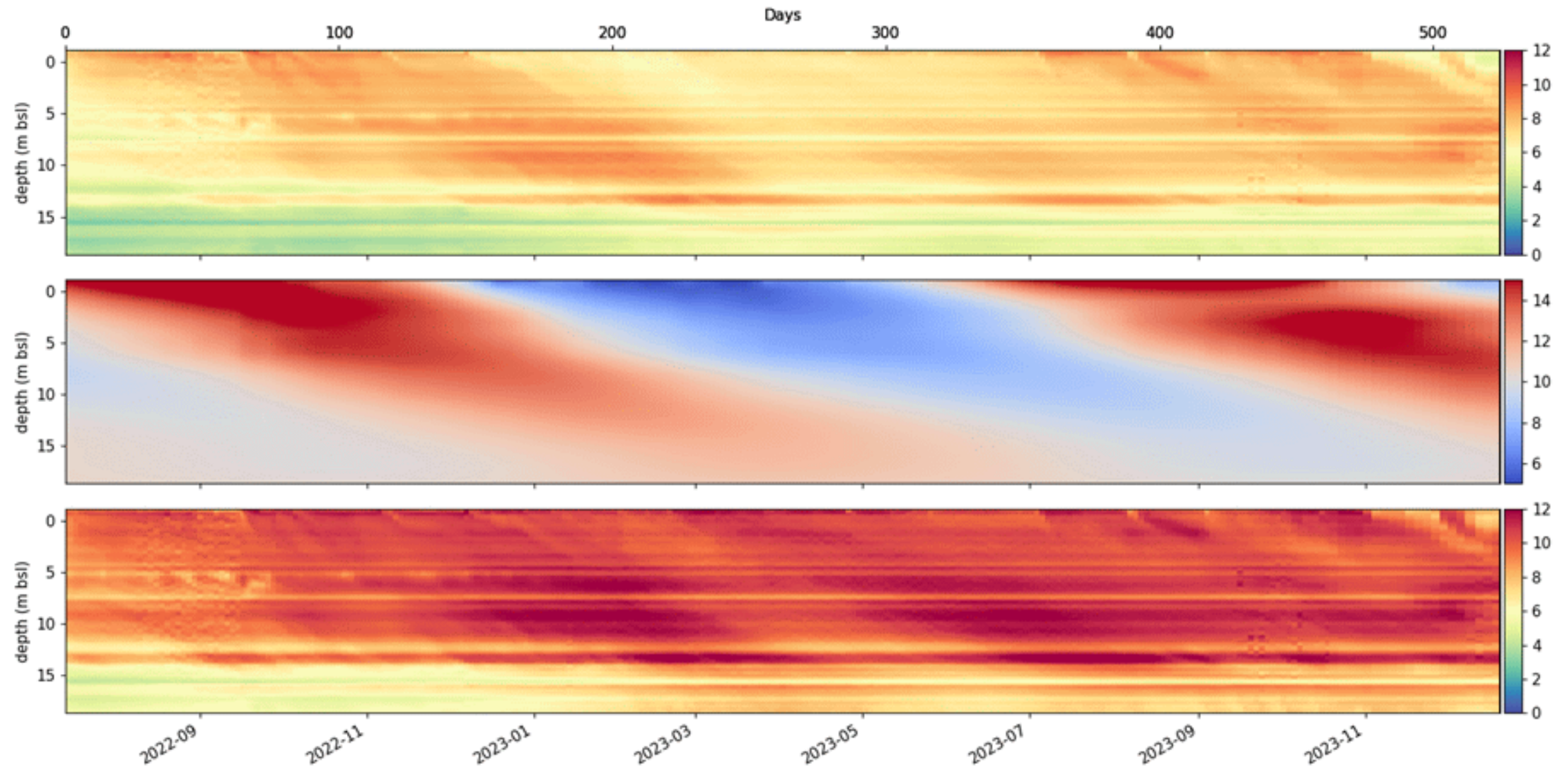


Temperature distribution crosshole ERT

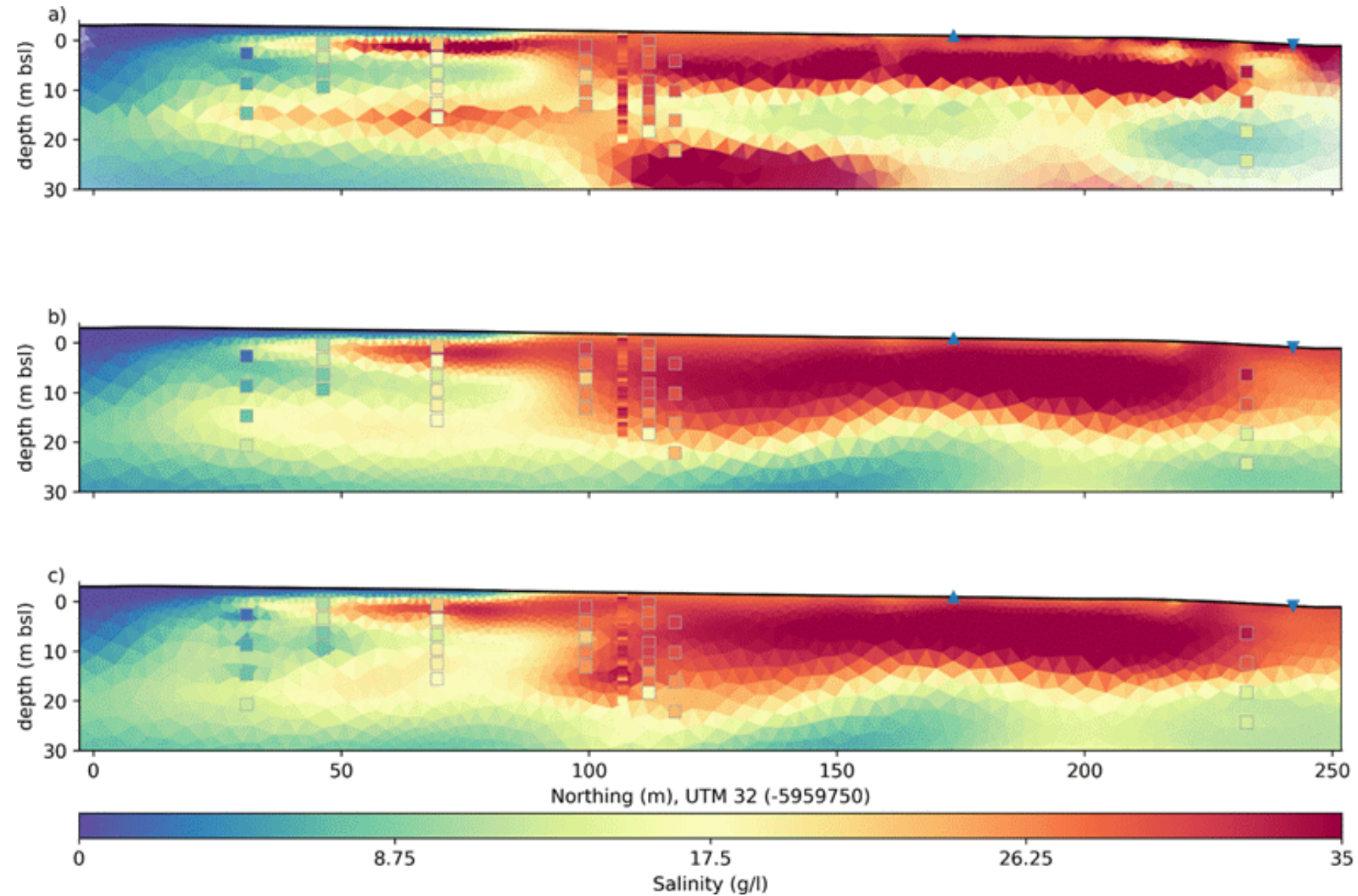


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# The salinity case (Dynadeep) - Temperature



# The salinity case - Salinity inversion



# Image appraisal

<https://gelmon25.pygimli.org>

# 🎯 Goals of this part of the workshop

- Get familiar with py**GIMLi**'s quality assessment tools
- Plot and examine sensitivities of single four-point configurations
- Utilize evaluation parameters as “cutoff” for the visualization
- Visualize image quality with depth (1D)
- **Advanced section:** Depth of Investigation (Oldenburg and Li, 1999)

## Basic terminology: Jacobian matrix:

- For a set of transform equations: , the Jacobian matrix is defined as:
- measure of the change in the  $i$ th data point as the  $k$ th parameter is changed
- measure of how strongly data depend on a parameter -> sensitivity

| <i>Parameter index <math>k</math></i>   |           |           |     |           |
|-----------------------------------------|-----------|-----------|-----|-----------|
| <i>Measurement index <math>i</math></i> | $J_{1,1}$ |           |     |           |
|                                         |           | $J_{2,2}$ |     |           |
|                                         |           |           | ... |           |
|                                         |           |           |     | $J_{i,k}$ |

## Basic terminology: Jacobian matrix:

- Single row: **sensitivity distribution of singular measurement** over whole model space
- Sum of single column: **sum of sensitivities** in a particular model cell (for all measurements of the dataset)
- Starting point for **sensitivity-based quality assessment methods**

| <i>Parameter index <math>k</math></i>   |           |           |     |           |
|-----------------------------------------|-----------|-----------|-----|-----------|
| <i>Measurement index <math>i</math></i> | $J_{1,1}$ |           |     |           |
|                                         |           | $J_{2,2}$ |     |           |
|                                         |           |           | ... |           |
|                                         |           |           |     | $J_{i,k}$ |

## Basic terminology: Depth of Investigation:

- generally refers to the depth of the model space, **“below which surface data are insensitive to the value of the physical property”**
- **interpretation of features below this depth should be handled with care**, since they might not be geologically feasible
- if we consider two inversion results that are carried out with **two different, homogeneous reference models** and their inverse models and , the DOI is computed by:
- **R will approach 0** where the **data constrain the model** -> inversion does not significantly depend on the defined reference model, inverse model provides **reliable information in those model area**

Oldenburg, D. W., & Li, Y. (1999). Estimating depth of investigation in DC resistivity and IP surveys. *Geophysics*. Society of Exploration Geophysicists.



## ▶▶ Happy hacking!

1. Extract and visualize single sensitivities
2. Plot coverage
3. Use coverage to adapt visualization inverse model
4. Examine image quality with depth at a specific location
5. Implement the DOI according to Oldenburg and Li (1999)

## 🚩 Takeaway messages:

- We visualized the **sensitivities of single four point arrays** on the model space and underlined **differences in the distribution** based on the electrode and dipole spacing. This helps to get a feeling of which configurations provide information on which part of the model space.
- We introduced the coverage as tool to assess the **spatial distribution of the sensitivity of all measurements across the model space**, which allows to **objectively evaluate the reliability** in different parts of the model.
- We build a tool that allows us to **cut out areas** from the result plot that hold **low coverage**. However, this **tool is method-independent** and can therefore also be utilized with other quality evaluation parameters, such as the Depth of Investigation (DOI) and the model resolution.
- We constructed a tool that allows us to assess the quality in a 1D column anywhere on the model. This way, we selectively evaluate the model quality for a specific region on the model space.

# References

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